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Secure & Scalable Static Website Hosting on AWS

Instructions: In today's digital landscape, businesses and individuals require reliable, secure, and globally accessible web platforms to establish their online presence. Traditional hosting solutions often involve complex infrastructure management and higher costs, making them less efficient for static websites.

This project, **"Secure & Scalable Static Website Hosting on AWS"**, addresses these challenges by leveraging Amazon Web Services to deliver a cost-effective, highly available, and secure hosting solution. Using **Amazon S3, Route 53, AWS Certificate Manager, and Amazon CloudFront**, the system provides end-to-end infrastructure for hosting static content with custom domain integration, DNS management, and global content delivery.

By implementing SSL/TLS certificates and CloudFront caching policies, the project ensures secure communication, compliance with modern standards, and optimized performance across regions. The result is a professional-grade static website that demonstrates practical skills in cloud infrastructure setup, domain management, and performance optimization.

The project demonstrates:

- Import a domain name on AWS
- Create a hosted zone and manage its DNS records;
- Create S3 Buckets to host website pages;
- Configure Route 53 to link the domain name to the website;
- Create an SSL certificate for the website using AWS Certificate Manager (ACM);
- Create CloudFront distributions to secure the website in HTTPS.

Tools & Technologies Used:

- **Amazon S3 –** Static website hosting and storage.
- **Amazon Route 53 –** DNS management and domain integration.
- **AWS Certificate Manager (ACM) –** SSL/TLS certificate generation.

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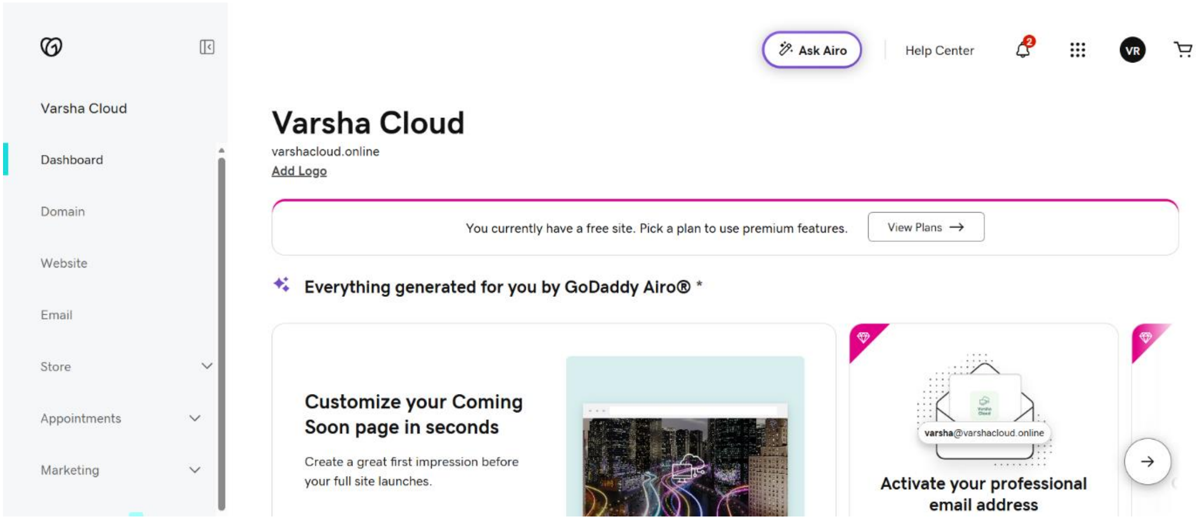
- **Amazon CloudFront –** Content delivery network (CDN) for global distribution.

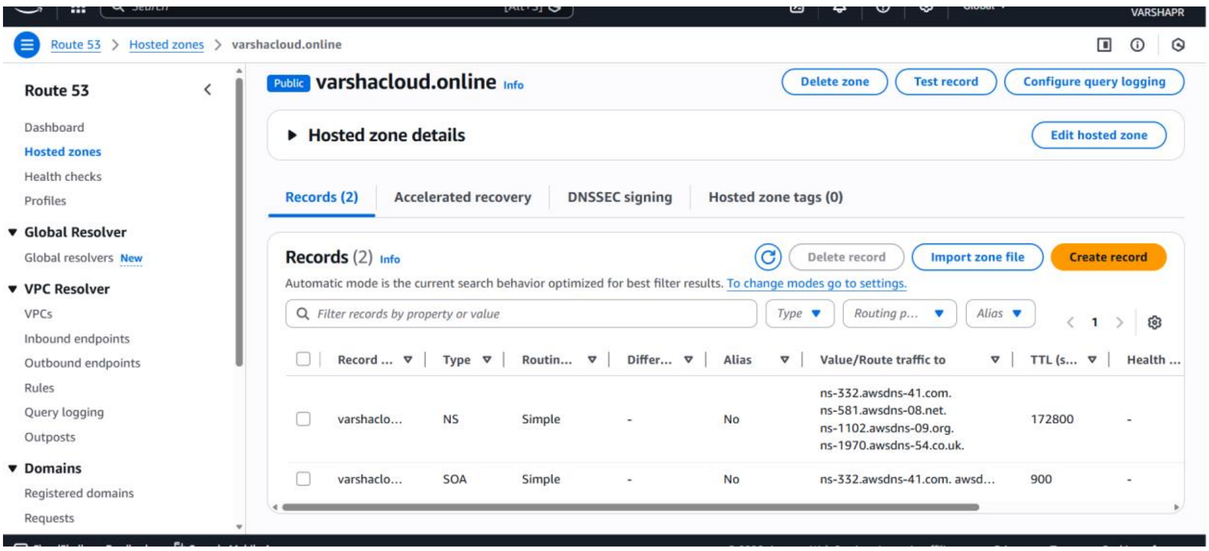
Use a domain name:



- Registered a custom domain to establish a professional online presence.
- Ensured domain availability and ownership through a registrar.

Create hosted Zone:

First, create your hosted zone, where you specify your domain name, in the example *cif-project.com*; then specify that you want your hosted zone to be public, otherwise it will be inaccessible from the public internet:



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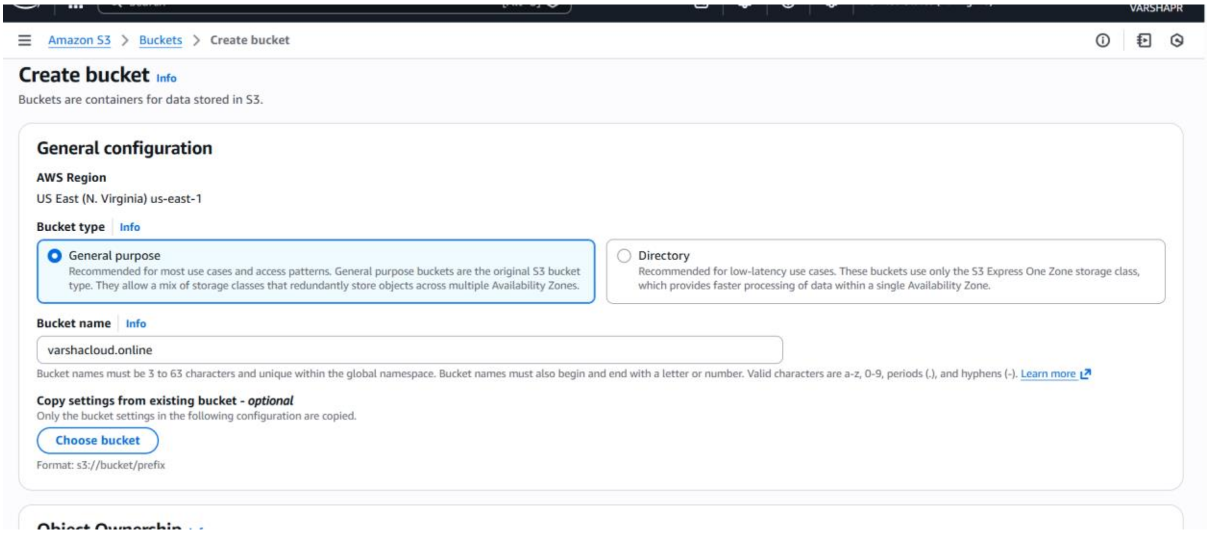
Your hosted zone is now created, and contains DNS records of type NS (Name Server) and SOA (Start of Authority).

Now you need to link your domain name purchased from a third-party domain registrar with your hosted zone. To do this, you will need to map the NS records of your hosted zone to the DNS configuration of the domain purchased from the domain registrar. The DNS configuration of the purchased domain can be easily found in the domain registrar's dashboard.

Create and Configure S3 Buckets:

To host your static website, you will need to create an S3 Bucket containing the pages of your website, then you will put the Bucket in public to make it accessible to all.

To begin with, create the main S3 Bucket which must have the same name as the website you are hosting:



Then allow this Bucket to be publicly available on the Internet:

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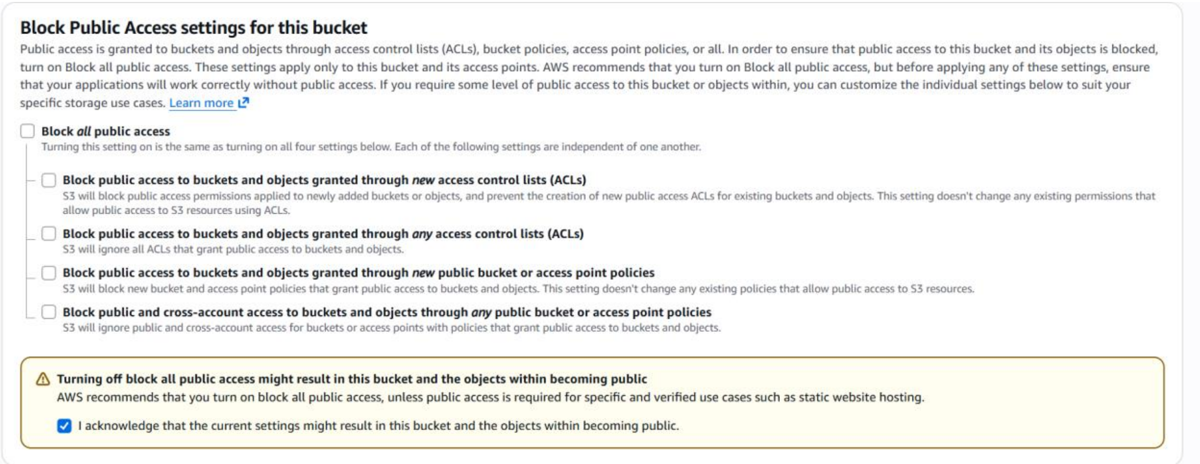
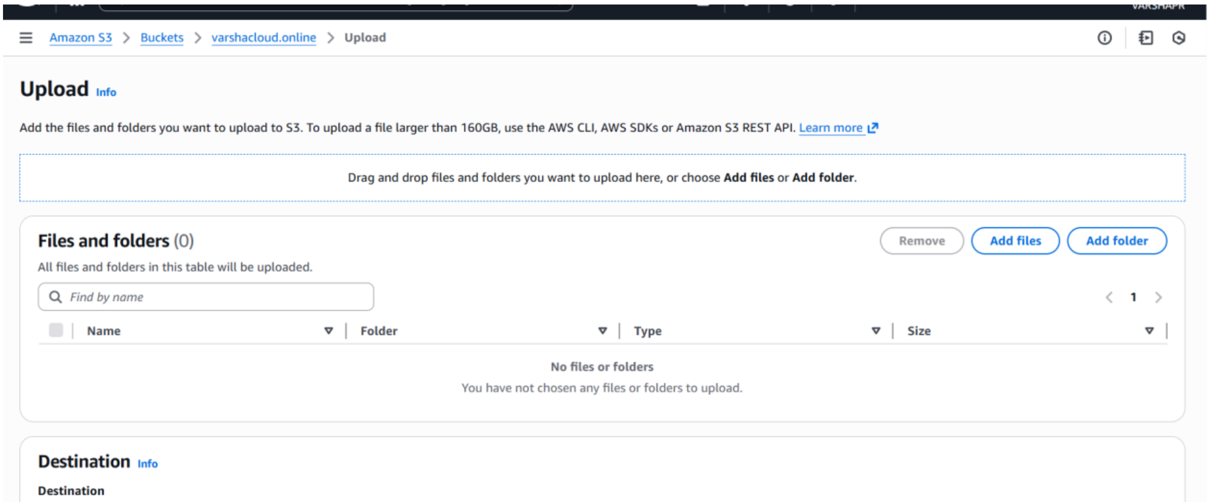


The main Bucket is now created, it is time to install the files of your website. To do this, go to your Bucket and click on Upload:



Add previously downloaded *index.html* and *error.html* files. The files are now objects in the main S3 Bucket:

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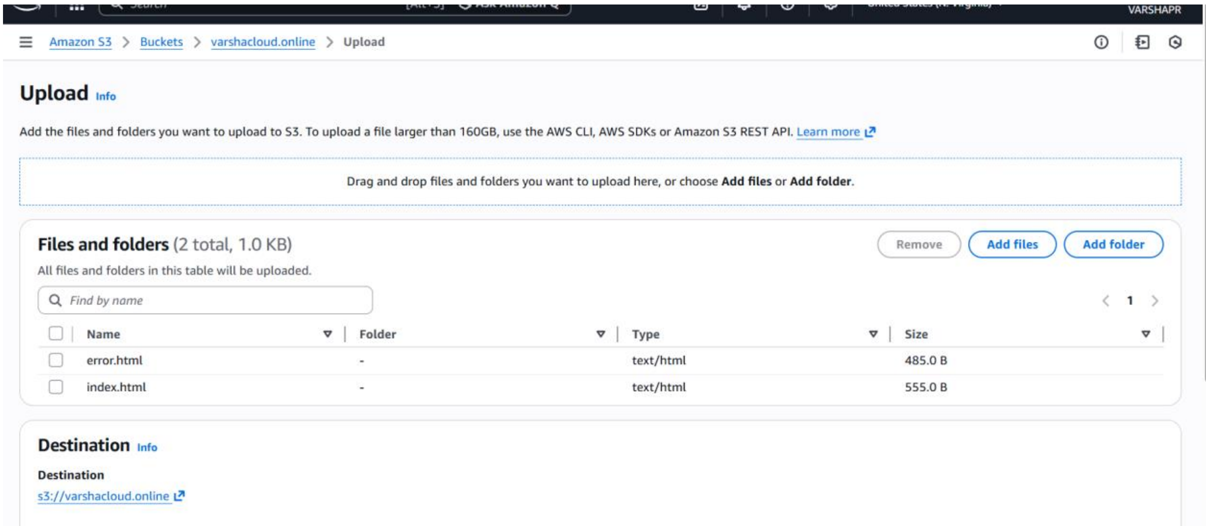
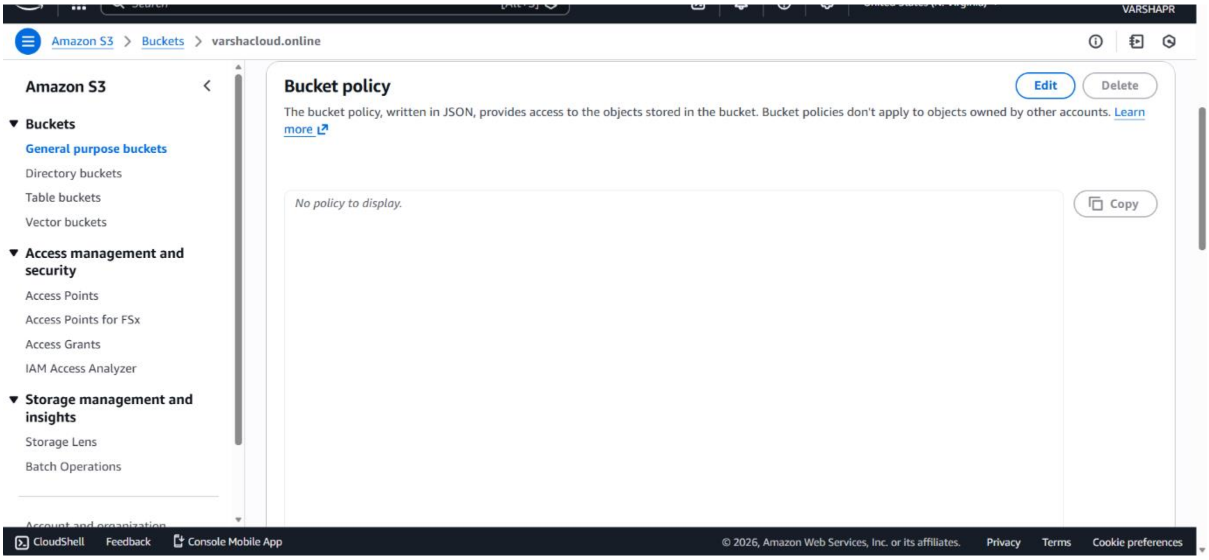


Move to **Permissions** to modify the Bucket Policy:

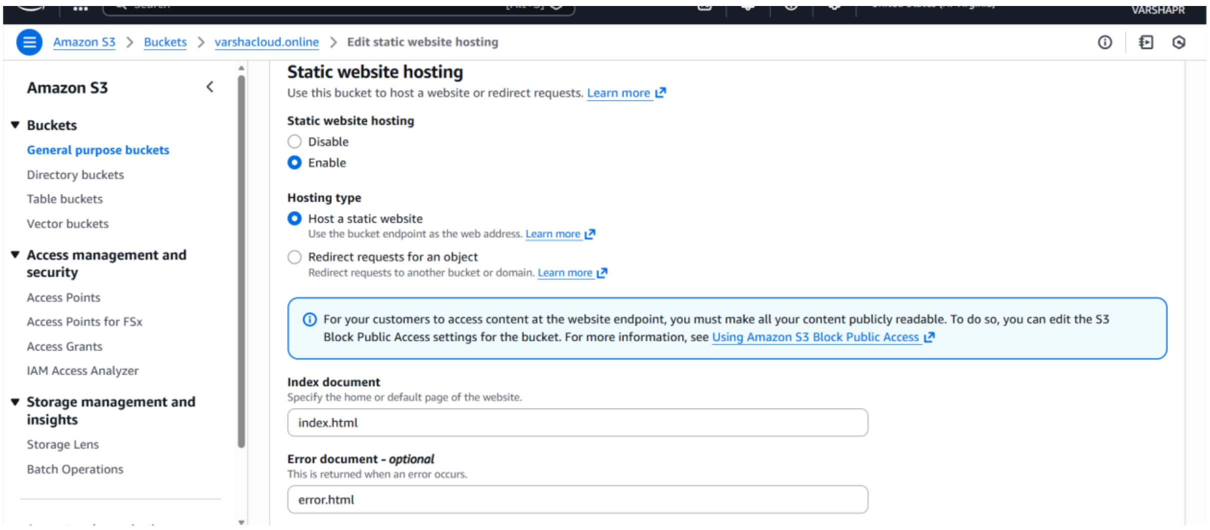


Insert the following policy and modify it slightly before validating:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicReadGetObject",
```

```
        "Effect": "Allow",
        "Principal": "*",
        "Action": [
          "s3:GetObject"
        ],
        "Resource": [
          "arn:aws:s3:::bucket_name/*"
        ]
      }
   ]
}
```

Last step: activate static web hosting by going to the **Static website hosting section**, at the bottom of the **Properties tab**:



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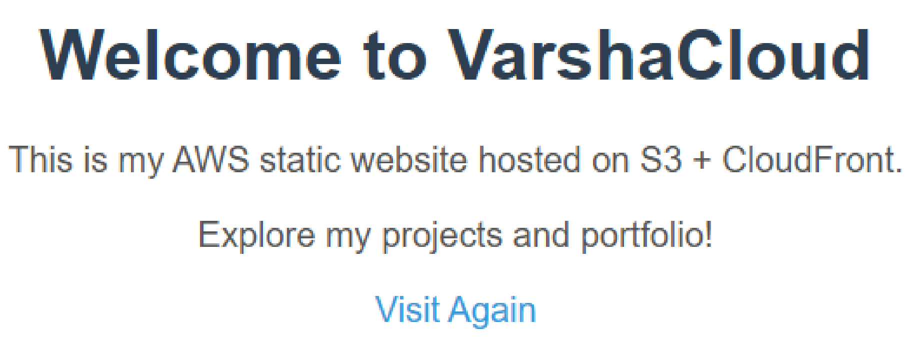
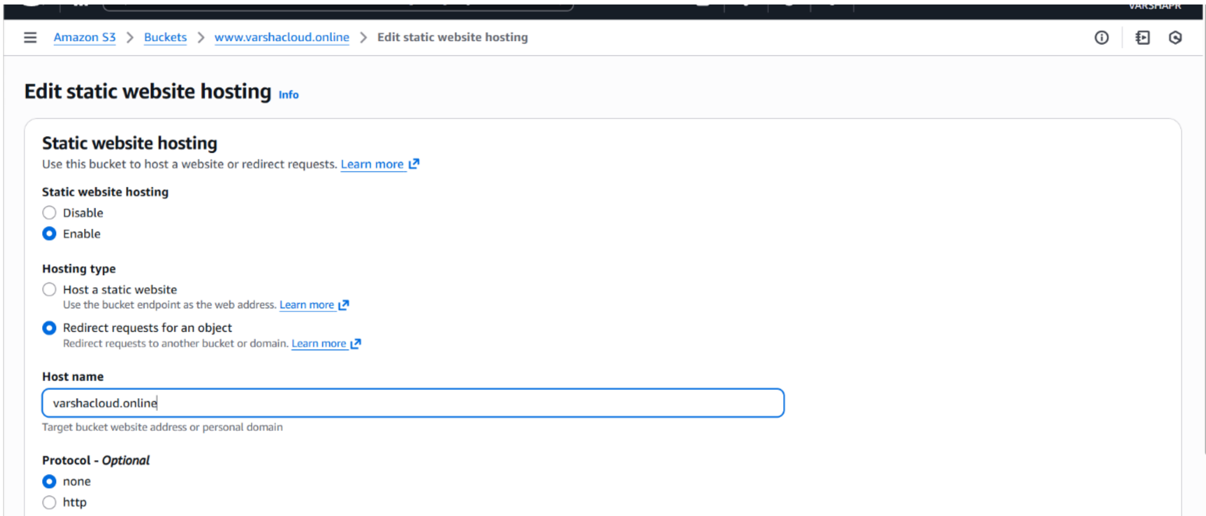
After saving changes, all you need to do is check that the site is accessible from a public address. Go back to the Static website hosting section to get the temporary URL of the S3 Bucket:

Now let's see what the S3 Bucket displays:



create the redirect S3 Bucket that will redirect all the traffic it receives to the main S3 Bucket. Then, go to the **Bucket configuration** > **Properties tab** > **Static website hosting section**:

ctivate the static website hosting, but this time by redirecting the requests. Specify the target Bucket , and leave the **Protocol field** as it is for the moment :



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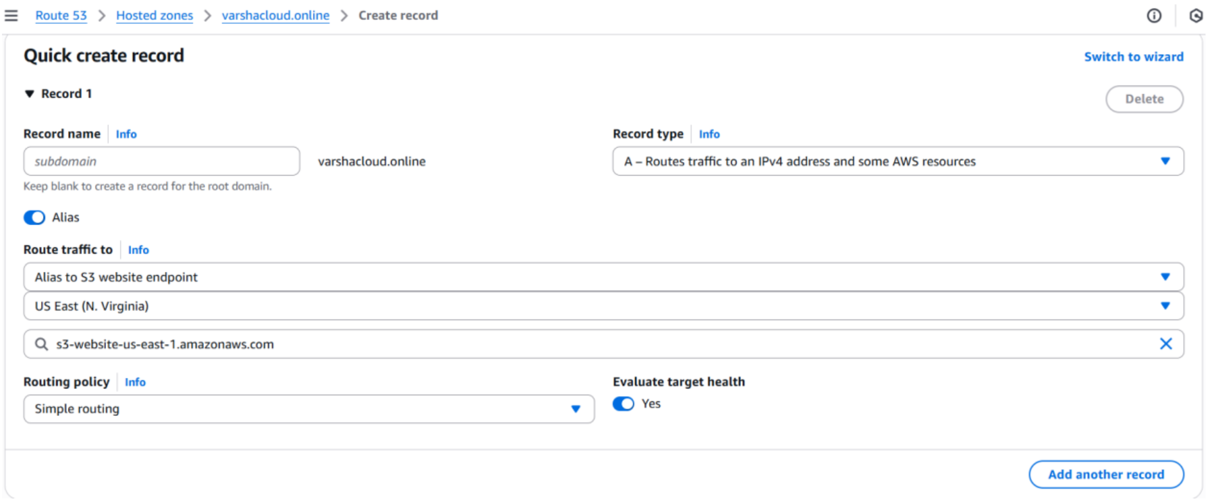
Link the domain to S3 Buckets with Route 53:

You want to route traffic from your main web address :



- If your main address is www, then add *www* to the record name. If your main address is non-www, leave the record name as it is;
- Select the Record type A to route the traffic to the main S3 Bucket;
- Activate the Alias to specify the S3 Bucket to which you want to route the traffic;
- Choose an Alias to S3 website endpoint in the region where the Bucket is located (here *us-east-1*), and select the Bucket that is automatically displayed in the selection;
- Routing policy will be simple routing;
- No need to evaluate the target health since the website is hosted on S3.

  Once the first record is created, wait several minutes before checking if your website is accessible via the main web address:

  You want to route the traffic of your redirect web address:

- If your redirect address is www, then add www to the record name. If your redirect address is non-www, leave the record name as it is;
- Select again a Record type in A to route the traffic to the redirect S3 Bucket;
- Activate the Alias to specify the S3 Bucket to which you want to route the traffic;
- Choose an Alias to S3 website endpoint in the region where the Bucket is located (here *us-east-1*), and select the Bucket that is automatically displayed in the selection;

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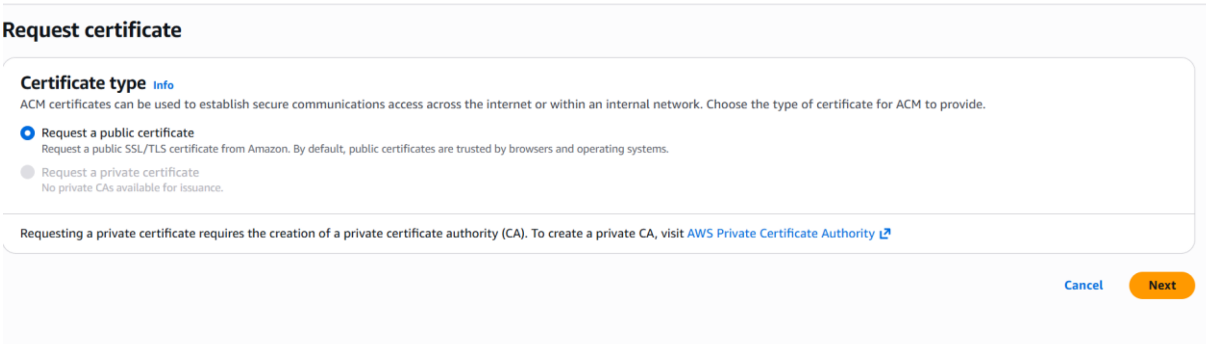
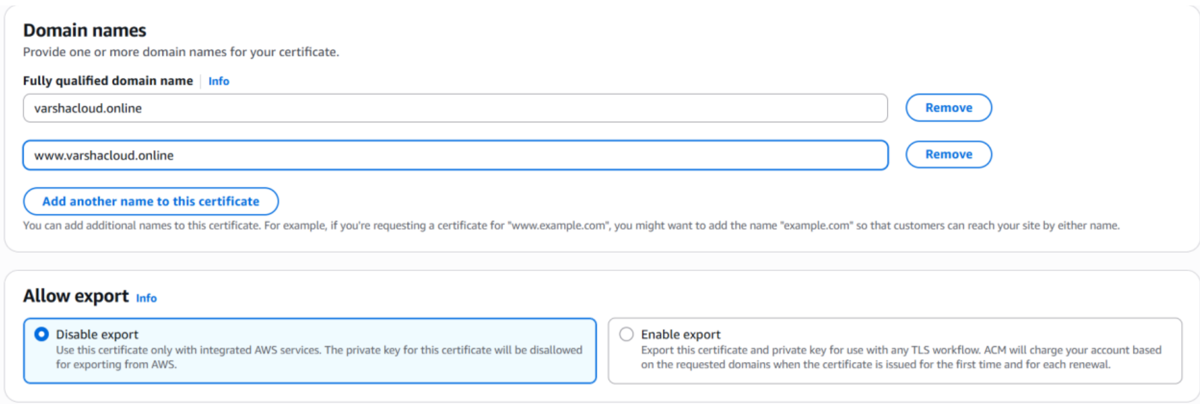
- Routing policy will be simple routing;
- No need to evaluate the target health since the site is hosted on S3.

Create SSL Certificate with AWS Certificate Manager:

To create a SSL certificate for your website, first go to AWS Certificate Manager and request a certificate. The certificate will be used on a public website, so you need to **request a public certificate**:



After that, all you have to do is validate the SSL Certificate creation request:



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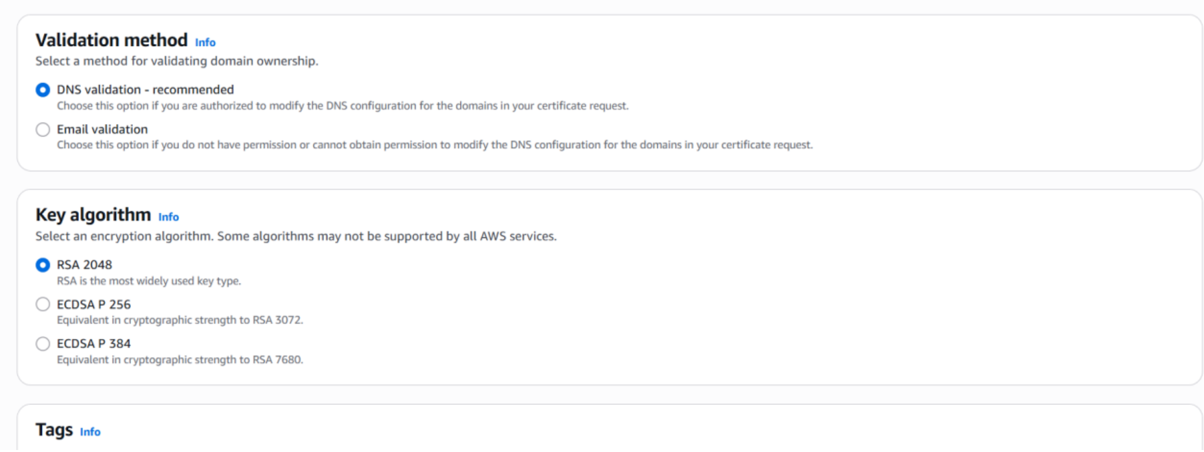
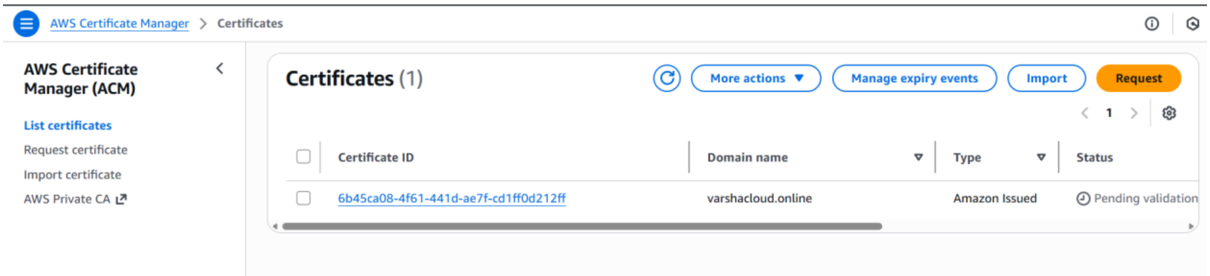


The certificate is created, but not yet validated; let's check its configuration:



In the configuration of the certificate, you need to add two DNS records of type CNAME in your hosted zone. Lucky you, AWS takes care of this: all you have to do is, first, select **Create records in Route 53**:

Then select the two FQDNs and finally **Create records**:

All you have to do now is wait a few minutes until you see the **Success status**:

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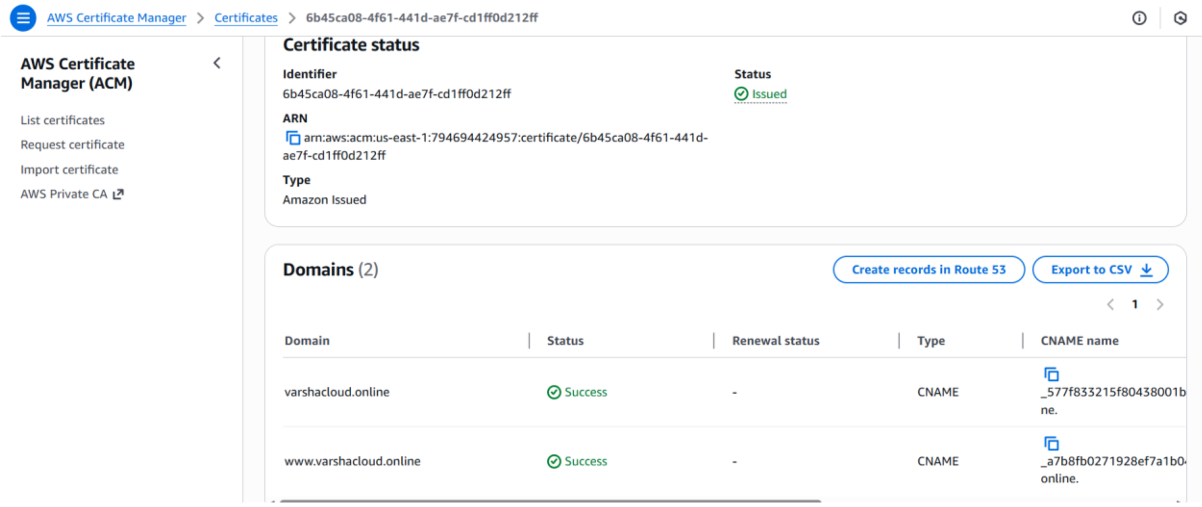
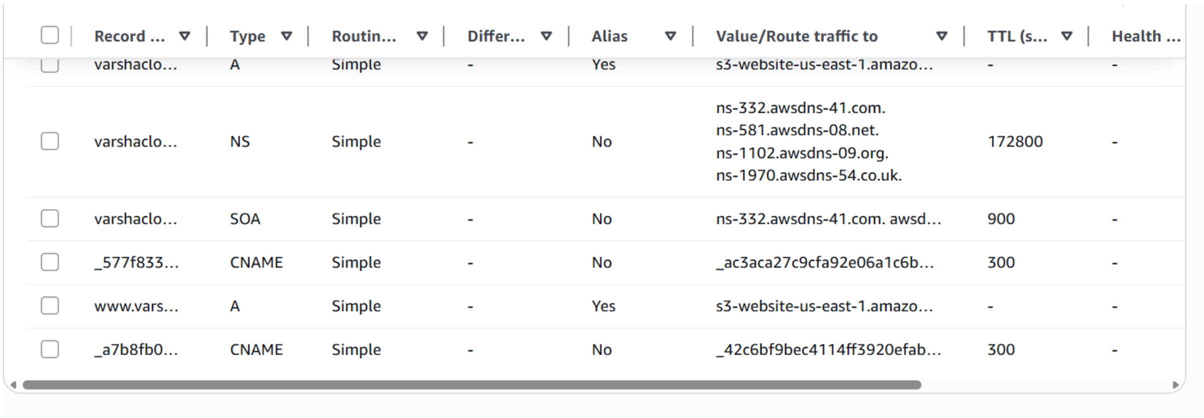


It's done, AWS has filled in the CNAME records for you. You can verify it by moving to your hosted zone:



Record ...	Type	Routin...	Differ...	Alias	Value/Route traffic to	TTL (s...	Health ...
varshaclo...	A	Simple	-	Yes	s3-website-us-east-1.amazo...	-	-
varshaclo...	NS	Simple	-	No	ns-332.awsdns-41.com. ns-581.awsdns-08.net. ns-1102.awsdns-09.org. ns-1970.awsdns-54.co.uk.	172800	-
varshaclo...	SOA	Simple	-	No	ns-332.awsdns-41.com. awsd...	900	-
_577f833...	CNAME	Simple	-	No	_ac3aca27c9cfa92e06a1c6b...	300	-
www.vars...	A	Simple	-	Yes	s3-website-us-east-1.amazo...	-	-
_a7b8fb0...	CNAME	Simple	-	No	_42c6bf9bec4114ff3920efab...	300	-

Create and Configure Cloud Distributions:

By using CloudFront, you will be able to deploy your website in HTTPS, and get other benefits as using Edge locations to provide faster access to the website for the users.

Before creating your main CloudFront distribution, go to the main S3 Bucket, and copy the Bucket URL into **Properties tab**, **static website hosting section**:

Now, create your CloudFront distribution. Paste the Bucket URL you previously copied in Domain field:

Main CloudFront Distribution:

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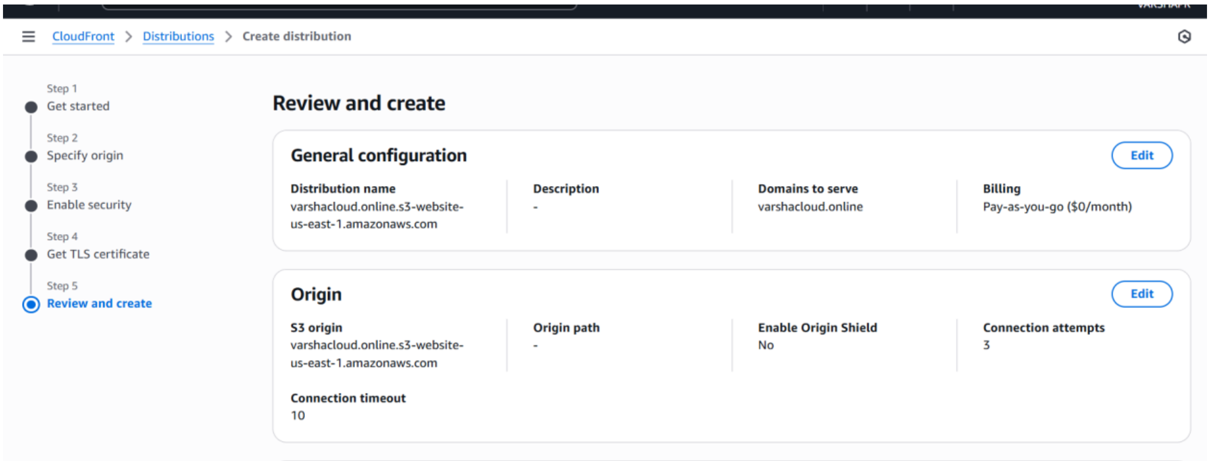
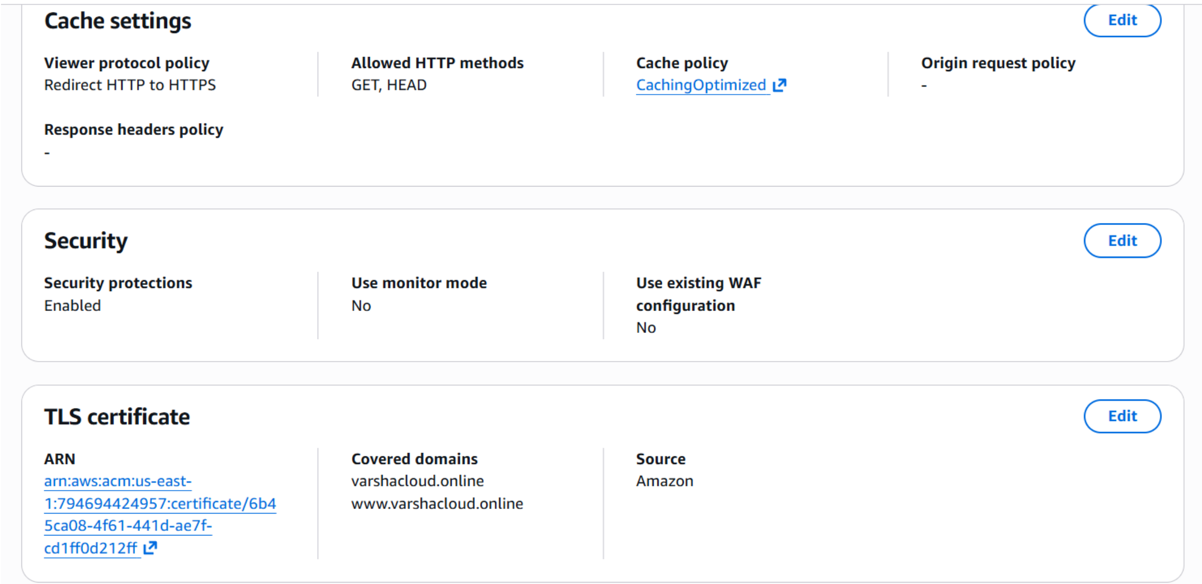
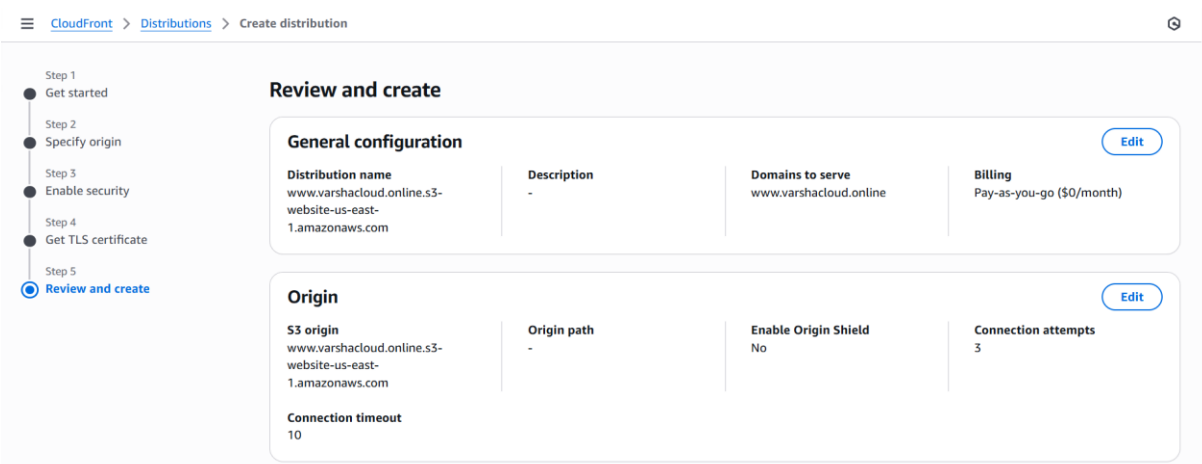




Redirect CloudFront Distribution:



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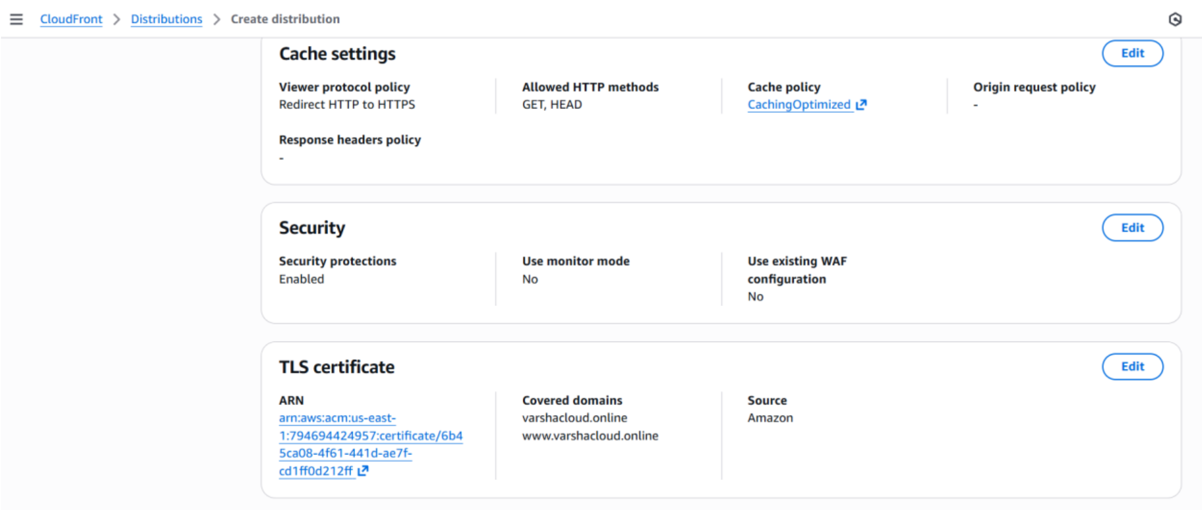
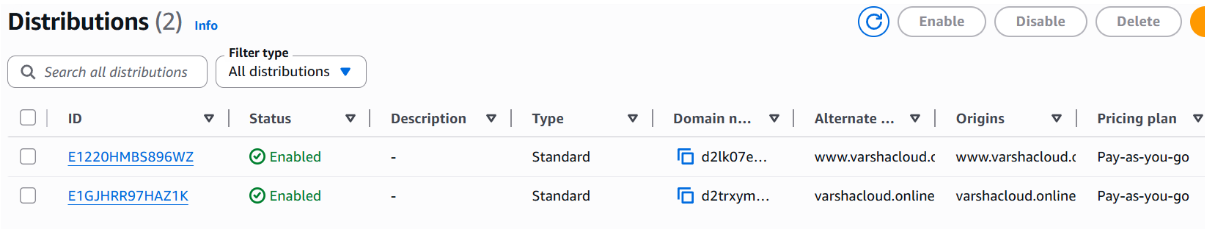
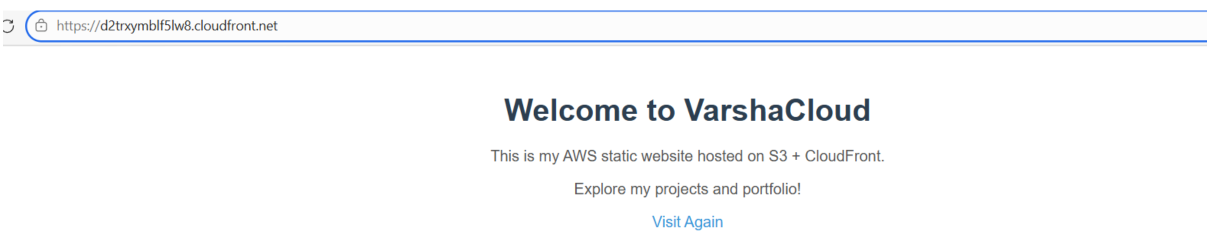


Go back to CloudFront, where you copy the web addresses of your CloudFront distributions (**Domain name field**) in the navigator search bar, in order to verify that you can access your site through these web addresses:



Start by copying the URL of the redirect CloudFront distribution:



Check with Main CloudFront Distribution.

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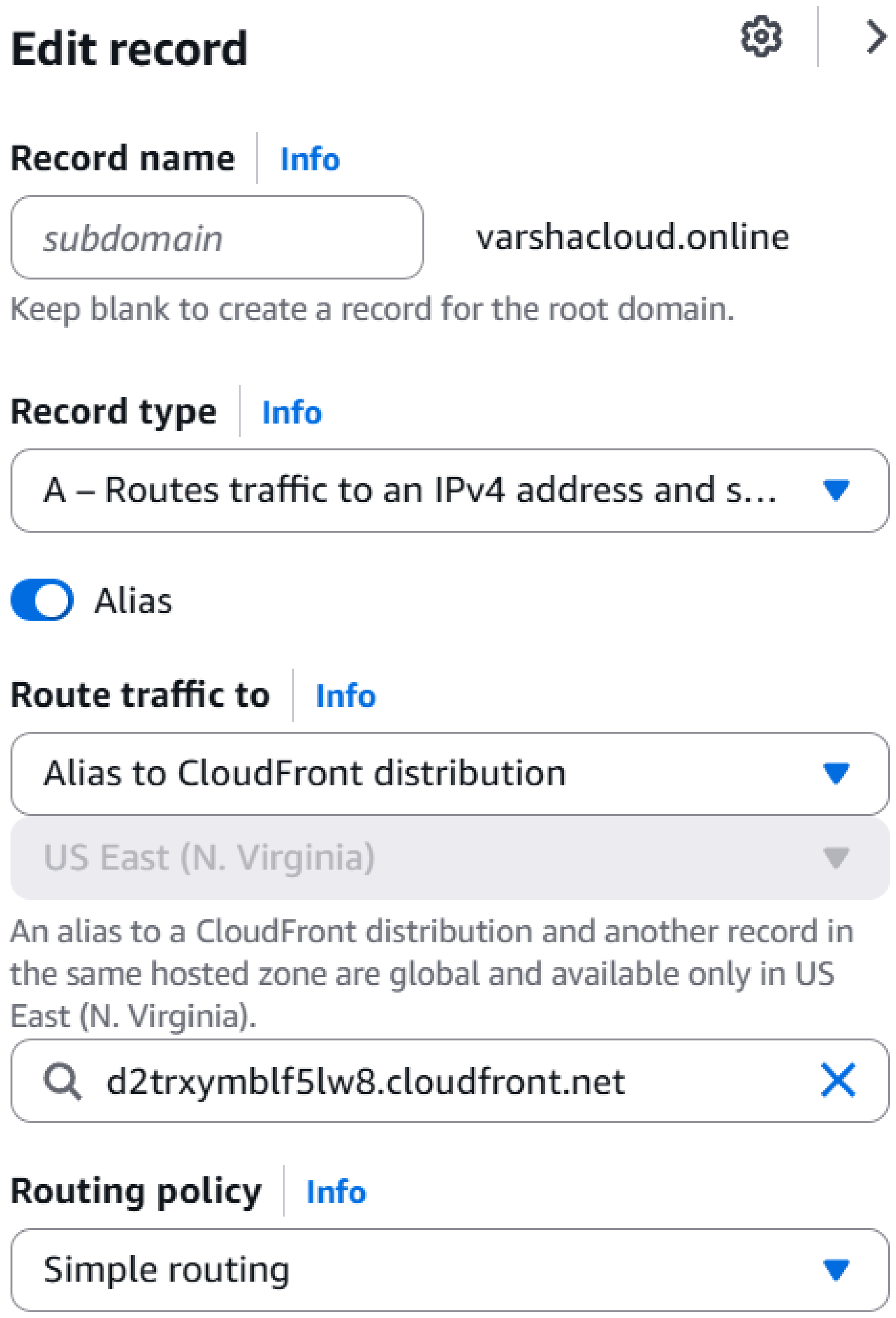
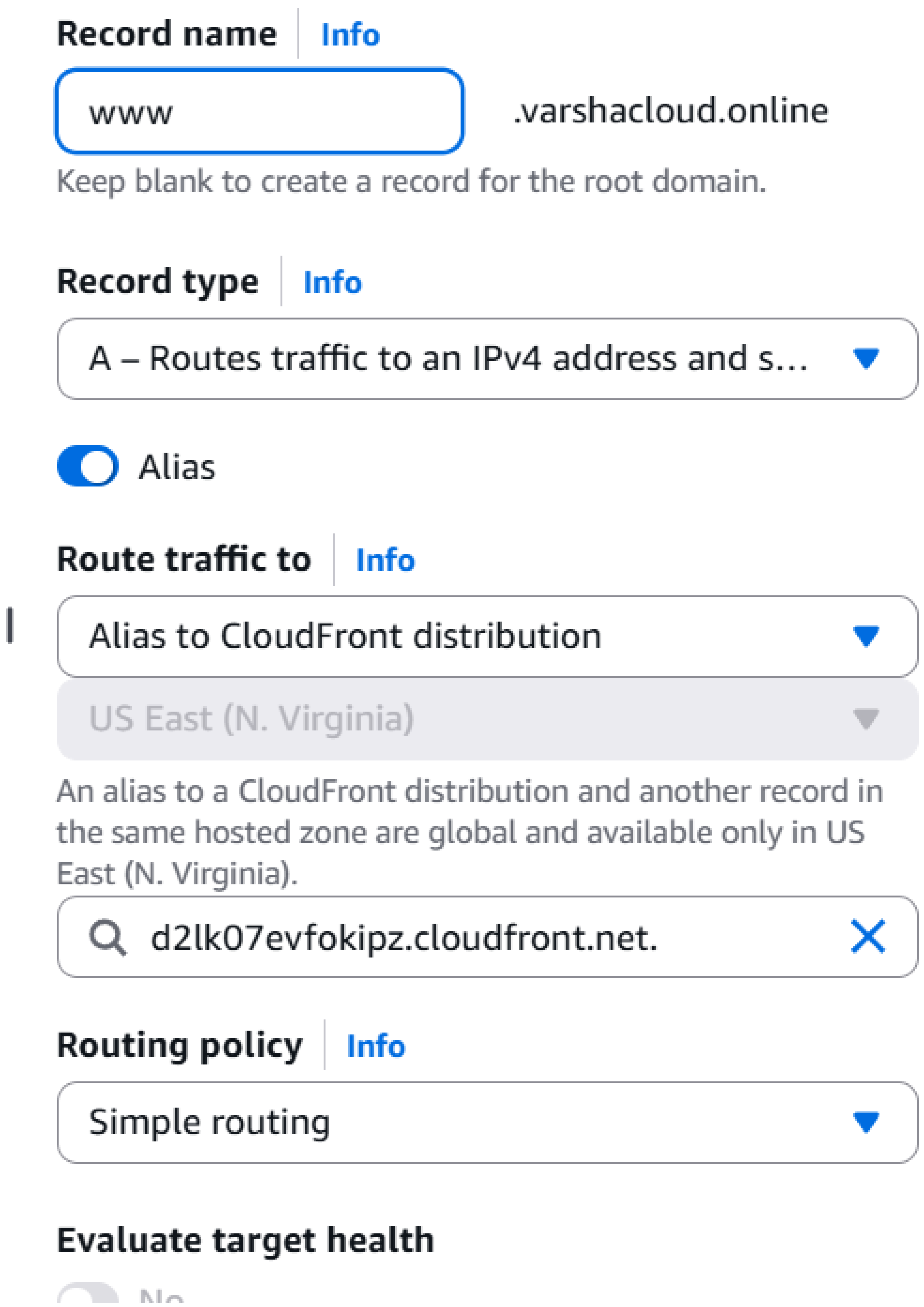
To finalize, head to Route 53 to make changes to the A records, so that the web addresses route traffic to your freshly created Cloudfront distributions.

The record for the main domain name should look like this:



And the record for the redirect domain name should look like that:



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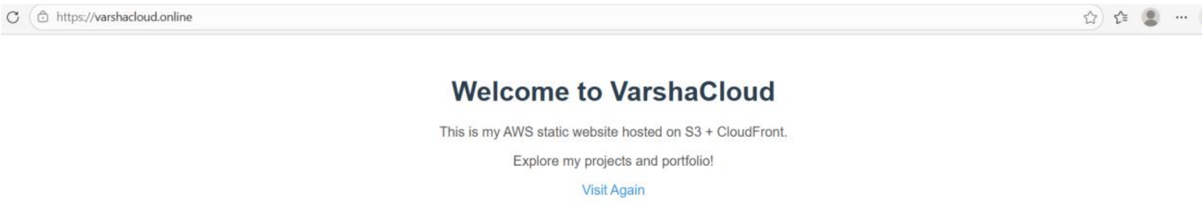
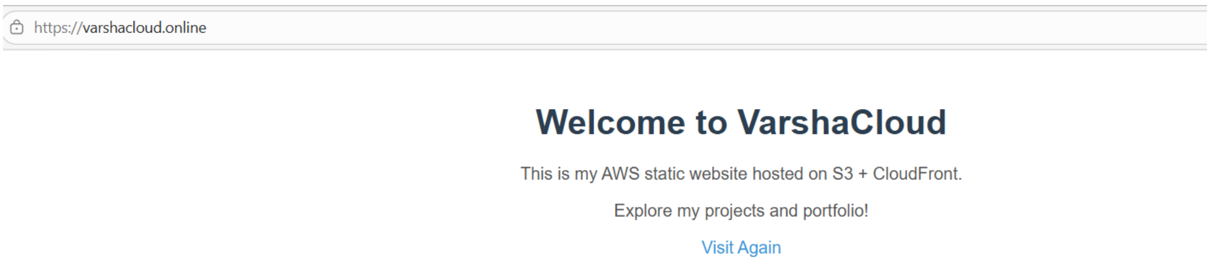
Let's check out the main and redirect domain in HTTP (http://varshacloud.online)(http://www.varshacloud.online):





By following this step we will host a secure and scalable website on aws.

Outcome

- Successfully deployed a **secure, globally accessible static website** using AWS services.
- The website is hosted on **Amazon S3** with a custom domain (varshacloud.online).
- **Route 53** DNS records correctly link the domain to the hosting bucket.
- **SSL/TLS certificate** from AWS Certificate Manager ensures encrypted HTTPS communication.
- **CloudFront distribution** delivers content with low latency and optimized performance worldwide.
- Demonstrated practical skills in:
  - Cloud infrastructure setup (S3, Route 53, CloudFront, ACM).
  - Domain integration and DNS management.

- Security implementation with SSL/TLS.
- Performance optimization using CDN caching.

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